Project Exercises

Due Date: April 29, 2026

Purpose

This assignment asks you to carry out the full data pipeline for an empirical research project: collect raw data from a primary source, clean and reshape it, construct analysis variables, run a validation regression, and produce a publication-quality figure. This is the daily work of empirical research—and the part that takes the most time.

You will also practice the professional workflow skills that make research reproducible: organized directory structures, version control with Git and GitHub, and compiled documents in LaTeX.

Choose Your Path

Path 1: Your Own Project. Use the data and research question from your project proposal. Demonstrate the collect → clean → visualize pipeline with your own data. You must download or access data from a primary source, clean and construct at least one analysis variable. Your pipeline must incorporate at least two of the following three data wrangling operations: crosswalking/merging, aggregation/collapsing, and reshaping. Produce at least one figure and one regression table that are automated/draft ready. Your write-up should describe what you did at each stage and what you learned about your data.

Path 2: Guided Exercise. Construct an immigration shift-share (Bartik) instrumental variable using U.S. Census microdata from IPUMS, following the approach of Kim, Merritt, and Peri (NBER Working Paper 32526). Details on the next page.

Both paths are graded on the same rubric. Path 2 provides more structure, but may be more complex; Path 1 requires more initiative.

Grading: The Project Proposal and Project Exercises together are worth 30% of your final grade, split **2:1 in your favor**—whichever you score higher on counts for 20%, the other for 10%.

Requirements (Both Paths)

1. **Directory structure.** Organize your project into a sensible folder structure. A suggested layout:

```
project/
  data/raw/          <- original downloaded data (never modify)
  data/clean/        <- processed datasets
  code/              <- your .do, .py, or .R files
  output/figures/    <- figures
  output/tables/     <- tables
  docs/              <- your write-up (.tex and .pdf)
  README.md          <- brief description of the project
```

You may adjust this structure, but your files must be logically organized—not dumped in one folder.

2. **GitHub.** Your project must live in a GitHub repository with a meaningful commit history. I should see multiple commits reflecting your progress—not a single commit at the end. Include a `README.md` that describes the project and explains how to reproduce your results.

3. **LaTeX.** Your write-up must be compiled in LaTeX (or R Markdown/Quarto with PDF output). Submit both the source file and the compiled PDF.

4. **Reproducibility.** Your code must run from start to finish without errors, given the raw data and any provided crosswalk files. Submit a log file (Stata) or console output documenting the complete run.

What to Submit (on Canvas)

- A link to your GitHub repository
- Your compiled PDF write-up (1–3 pages, excluding figures and tables)
- Your code file(s) and log file
- At least one figure and one regression table

Collaboration & AI Policy

Same policy as problem sets. You may use AI tools for debugging and code assistance, but you must document your use. Include an AI log or attest to non-use in your write-up.

Guided Exercise: Constructing an Immigration Shift-Share IV

Your Goal

Construct a shift-share (Bartik) instrumental variable for immigration across U.S. commuting zones, using five decades of Census microdata (1970–2010). Then use the instrument to estimate the causal effect of immigration on local wages.

A key part of this exercise is **planning your own code**. You are given the formula, the data sources, and the deliverables—but you must figure out the pipeline steps yourself. Think carefully about what operations are needed, in what order, and at what unit of observation. Sketch your plan before you start coding.

Background

Immigrants from a given origin country tend to settle where earlier immigrants from the same country already live—driven by established ethnic networks, not only current local labor demand. A shift-share instrument exploits this pattern. The instrument for commuting zone c in decade t is:

$$\widetilde{B}_{c,t} \;=\; \frac{1}{\text{pop}_{c,1970}} \sum_{o} \underbrace{\frac{\text{imm}_{c,o,1970}}{\text{imm}^{\text{national}}_{o,1970}}}_{\text{1970 share}} \times \underbrace{\left(\text{imm}^{\text{national}}_{o,t} - \text{imm}^{\text{national}}_{o,t-1}\right)}_{\text{national change, decade } t}$$

A detailed description of the methodology, notation, and identification logic is provided in the attached reference document (`kmp_imm_shift_share_excerpt.tex`).

Reference: Kim, Gueyon, Cassandra Merritt, and Giovanni Peri. "Measuring and Predicting 'New Work' in the United States: The Role of Local Factors and Global Shocks." NBER Working Paper No. 32526, May 2024.

Data Sources

1. **IPUMS USA** (`https://usa.ipums.org`). Download Census microdata for 1970, 1980, 1990, 2000, and 2010. You will need variables for geography, person weights, age, detailed birthplace (`BPLD`), detailed education (`EDUCD`), wage income (`INCWAGE`), hours and weeks worked (`UHRSWORK`, `WKSWORK1`), and group quarters (`GQ`). The geography variable differs by decade (county groups for 1970–1980; PUMAs for 1990–2010). Using the IPUMS API is expected.

2. **Dorn crosswalks** (`https://www.ddorn.net/data.htm`). Download the crosswalk files that map Census county groups and PUMAs to 1990-definition commuting zones. Each crosswalk contains an allocation factor for splitting geographic units across CZs.

3. **Birthplace crosswalk** (provided on Canvas: `cw_bpld_kmp.dta`). Maps IPUMS birthplace codes to ∼84 origin-country groups. Code 78 = U.S.-born; code 82 = other/unspecified (drop both from immigrant counts).

The Pipeline

A key part of this exercise is **planning your code**. Below is what you need to build and the intermediate datasets along the way. You must figure out the specific operations (merges, collapses, reshapes) and their order.

Stage 1: Collect.

- Download five IPUMS Census extracts (1970–2010) in Stata format. The geography variable differs by decade: `CNTYGP97` (1970), `CNTYGP98` (1980), `PUMA` (1990–2010). Include: `STATEFIP, PERWT, AGE, BPLD, EDUCD, INCWAGE, UHRSWORK, WKSWORK1, GQ`. Use `BPLD` not `BPL`; `EDUCD` not `EDUC`.
- Download four Dorn crosswalk files (one per geography vintage). Each has a different merge key and an allocation factor for splitting geographic units across CZs. The 2000 crosswalk covers 2010 as well.
- Using the IPUMS API (demonstrated in class) is recommended—five manual extracts is tedious.

Stage 2: Process each decade → CZ-level dataset. For each of the five Census decades, produce a commuting-zone-level dataset containing: total population, total immigrant count, average wage, and (for 1970) immigrant counts by origin group. Along the way:

- Convert Census geography to commuting zones using the appropriate Dorn crosswalk. Merge keys differ by decade—read the crosswalk documentation. Multiply person weights by the allocation factor to create adjusted weights.
- Classify individuals by origin using the birthplace crosswalk (`cw_bpld_kmp.dta`). Foreign-born = `kmp_ctry` $\neq 78$. Drop code 82 (other/unspecified).
- Apply sample restrictions: drop Alaska and Hawaii (`statefip` 2, 15); drop group quarters types 0, 3, 4; drop age < 16.
- Collapse from individual-level to CZ $\times$ origin, then to CZ-level aggregates.
- For each decade, also compute **national** immigrant totals by origin (sum across all CZs)—you need these to compute the shares and shifts.

Stage 3: Compute shares, shifts, and the shift-share instrument.

- **Shares** (from 1970): For each origin o, what fraction of its national immigrant population lives in CZ c? $\rightarrow$ one value per CZ $\times$ origin pair.
- **Shifts** (from national totals): For each origin o and decade t, what is the national change in immigrants? $\rightarrow$ one value per origin $\times$ decade.
- **Shift-share**: Multiply shares $\times$ shifts, sum across origins, normalize by 1970 CZ population $\rightarrow$ one instrument value per CZ $\times$ decade.

Stage 4: Build the panel.

- Merge the five CZ-level datasets and the shift-share instruments into a single CZ $\times$ decade panel.
- Compute actual immigration changes: $\Delta \text{imm}_{c,t} = (\text{imm}_{c,t} - \text{imm}_{c,t-1})/\text{pop}_{c,1970}$.
- Compute wage changes: $\Delta \text{wage}_{c,t} = \text{wage}_{c,t} - \text{wage}_{c,t-1}$.
- Reshape to long format (one row per CZ–decade).

Stage 5: Estimate and visualize.

- **Map:** Choropleth of the shift-share instrument across CZs (at least one decade). This is a useful diagnostic—does the predicted pattern make geographic sense?
- **First stage:** Regress $\Delta \text{imm}_{c,t}$ on $\widetilde{B}_{c,t}$ with year FE, clustered SEs. Scatter plot by decade.
- **Wage regressions:** OLS and IV/2SLS of $\Delta \text{wage}_{c,t}$ on $\Delta \text{imm}_{c,t}$, instrumenting with $\widetilde{B}_{c,t}$. Year FE, clustered SEs. Report both in a table.
- **Interpret:** Is the instrument strong? How do OLS and IV compare? What does the difference suggest about bias?

Hints

- Stages 2 and 3 are repetitive across five decades—I strongly encourage you to write a loop or function rather than copy-pasting code five times. Automation is rewarded in the rubric.
- The hardest part is the data construction, not the regressions. Think about what unit of observation you need at each stage and what operations get you there.
- The 1970 Census is special: it provides the base-year shares *and* the starting counts. All other decades provide actual counts and wages only.
- Read the Dorn crosswalk documentation—the merge keys and allocation factor names differ across decades.

Extensions (for Exceptional credit)

- **Coefficient plot** of regression results across specifications or decades.
- **Education split:** Separate instruments for high-ed (`EDUCD` $\geq$ 65) and low-ed (`EDUCD` $\leq$ 64) immigrants.
- **Additional maps** across decades or for actual immigration change.

Grading Rubric

This rubric applies to **both paths**. Each category is assessed on a Baseline / Strong / Exceptional scale. Baseline corresponds to a satisfactory submission; Strong to good work; Exceptional to excellent. Most students should aim for Strong across all categories.

Category	Weight	Performance Standards
Data collection	20%	**Baseline:** Data is downloaded from IPUMS (or appropriate primary source for Path 1) and present in the repository. **Strong:** Correct samples and variables selected for all five decades; extract choices documented in the write-up or README. **Exceptional:** Data collection automated via the IPUMS API; collection script is included and reproducible.
Data cleaning & wrangling	20%	**Baseline:** Code attempts the key operations—geography crosswalk merge, origin classification, sample restrictions, and collapse—for at least two decades. **Strong:** All five decades processed correctly; collapse produces the right unit of observation (CZ $\times$ origin, then CZ); shift-share instruments properly constructed and normalized for all decades. **Exceptional:** Multi-decade processing is automated (loops or functions rather than copy-paste); intermediate checks (observation counts, summary statistics) included; edge cases handled cleanly.
Estimation & visualization	20%	**Baseline:** A first-stage regression and a map of the instrument are produced. **Strong:** First-stage and IV wage regressions correctly specified with year FE and clustered SEs; map and scatterplots paneled by decade with clear labels. OLS vs. IV comparison interpreted in the write-up. **Exceptional:** Coefficient plot used to present regression results; interpretation demonstrates understanding of the instrument's economic logic and direction of OLS bias. Education split or additional specifications attempted.
Directory management & GitHub	10%	**Baseline:** Files are in a GitHub repository with at least one commit. **Strong:** Sensible folder structure (e.g., separate `data/`, `code/`, `output/` folders); multiple commits with informative messages; README describes the project. **Exceptional:** Commit history reflects a logical development workflow; repository is clean (no unnecessary files; `.gitignore` used appropriately).
LaTeX write-up	10%	**Baseline:** A compiled PDF is submitted with basic descriptions of the steps taken. **Strong:** Write-up is organized, concise, and includes properly formatted figures and tables with captions. Source `.tex` file is included. **Exceptional:** Write-up reads like a professional short research note; equations, table notes, and cross-references are polished.
Reproducibility	20%	**Baseline:** Code files are submitted. **Strong:** Code runs end-to-end without errors; a log file documenting the full run is submitted. **Exceptional:** Code includes a header listing dependencies, uses relative paths throughout, and is commented enough that a classmate could follow the logic without additional guidance.

Appendix: Detailed Step-by-Step Reference

The following provides detailed directions for each stage of the pipeline. Use this as a reference if you get stuck—but try to plan your approach before reading ahead.

The Immigration Shift-Share Instrument

Kim, Merritt, and Peri (2024) study how local exposure to global shocks—including immigration—affects the creation of "new work" (newly emerging job types) across U.S. commuting zones.[1] Because local immigration is endogenous to local labor-market conditions (immigrants may be drawn to booming areas), they construct a shift-share (Bartik) instrumental variable that predicts local immigration using only historical settlement patterns and national-level inflows.

Intuition. Immigrants from a given origin country tend to settle where earlier immigrants from the same country already live—driven by established ethnic networks, not current local demand. By combining each commuting zone's historical composition of immigrant origins with subsequent national immigration growth from those origins, the instrument predicts local immigration inflows that are driven by national trends rather than local economic conditions.

Construction. The instrument has two ingredients:

1. **Shares (from the 1970 Census).** For each commuting zone c and origin group o, compute:
$$s_{c,o,1970} = \frac{\text{Immigrants}_{c,o,1970}}{\text{Immigrants}^{\text{national}}_{o,1970}}$$
This is the fraction of origin o's total U.S. immigrant population that was living in commuting zone c in 1970.

2. **Shifts (national growth by origin).** For each origin group o and decade t, compute the national change in immigrant population:
$$\Delta M_{o,t} = \text{Immigrants}^{\text{national}}_{o,t} - \text{Immigrants}^{\text{national}}_{o,t-1}$$

The shift-share instrument for commuting zone c in decade t is then:
$$\widetilde{B}_{c,t} = \frac{1}{\text{pop}_{c,1970}} \sum_{o} s_{c,o,1970} \times \Delta M_{o,t}$$
where $\text{pop}_{c,1970}$ is the total population of commuting zone c in 1970 (used for normalization).

Identification logic. The instrument is valid if the 1970 settlement shares are uncorrelated with decade-t local labor-demand shocks (conditional on controls). This is plausible because the shares are determined decades before the outcome period, and immigrant location choices in 1970 reflect historical network formation rather than current economic conditions. Following the framework of Goldsmith-Pinkham, Sorkin, and Swift (2020), identification comes from the exogeneity of the historical shares.

[1] Kim, Gueyon, Cassandra Merritt, and Giovanni Peri. "Measuring and Predicting 'New Work' in the United States: The Role of Local Factors and Global Shocks." NBER Working Paper No. 32526, May 2024.

Data. The instrument is constructed from U.S. Decennial Census microdata obtained through IPUMS USA, covering the 1970, 1980, 1990, 2000, and 2010 Censuses. Individual-level data are mapped to commuting zones (1990 definitions) using crosswalks from Dorn (2009), and birthplace codes are mapped to approximately 80 origin-country groups using a custom crosswalk.

A.1 Data Collection

Register for a free account at https://usa.ipums.org. You need **five** Census extracts:

Decade	IPUMS Sample	Geography Var	Other Variables
1970	1970 1% Form 2 State	CNTYGP97	STATEFIP, PERWT, AGE, BPLD, EDUCD, INCWAGE, UHRSWORK, W
1980	1980 5% State	CNTYGP98	(same)
1990	1990 5% State	PUMA	(same)
2000	2000 5%	PUMA	(same)
2010	2010 ACS	PUMA	(same)

Download the Dorn crosswalks from https://www.ddorn.net/data.htm:

File	Merge key	Alloc. factor
cw_ctygrp1970_czone_corr.dta	cty_grp70	afact
cw_ctygrp1980_czone_corr.dta	$1000 \times$ statefip + cntygp98	afactor
cw_puma1990_czone.dta	$10000 \times$ statefip + puma	afactor
cw_puma2000_czone.dta	$10000 \times$ statefip + puma	afactor

The 2000 PUMA crosswalk is used for **both** 2000 and 2010.

A.2 Process 1970 Census

(a) **Convert to commuting zones.** Rename cntygp97 to cty_grp70 and merge (many-to-many, e.g. joinby) with the 1970 Dorn crosswalk. Create adjusted weights: aperwt = afact $\times$ perwt.

(b) **Classify by origin.** Merge cw_bpld_kmp.dta on bpld. Keep matched only.

(c) **Sample restrictions.** Drop AK/HI (statefip 2, 15). Drop GQ types 0, 3, 4. Drop age < 16.

(d) **Collapse** to CZ $\times$ origin (sum adjusted weights).

(e) **Drop** origin code 82.

(f) **CZ population:** Sum adjusted weights across all origins (including US-born, code 78) $\rightarrow$ pop1970.

(g) **National totals:** For each origin, sum immigrant counts across all CZs. Save separately.

(h) **Shares:** $s_{c,o} = \text{imm}_{c,o} \,/\, \text{imm}_o^{\text{national}}$.

(i) **CZ aggregates:** Sum across foreign-born origins $\rightarrow$ total immigrants. Compute average wage (weighted, excluding zero/missing INCWAGE). Save with pop1970, imm1970, wage1970.

A.3 Process 1980 Census

Same pipeline as A.2, but:

- Geography merge key: $1000 \times$ `statefip`$+$`cntygp98`. Crosswalk: `cw_ctygrp1980_czone_corr.dta`. Allocation factor: `afactor` (not `afact`).
- Save national totals by origin, CZ-level immigrants, population, and average wage.

A.4 Process 1990, 2000, 2010

Same pipeline. Geography differences only:

Decade	Merge key formula	Crosswalk file	Alloc. factor
1990	$10000 \times$ `statefip` $+$ `puma`	`cw_puma1990_czone.dta`	`afactor`
2000	$10000 \times$ `statefip` $+$ `puma`	`cw_puma2000_czone.dta`	`afactor`
2010	$10000 \times$ `statefip` $+$ `puma`	`cw_puma2000_czone.dta`	`afactor`

Save national totals by origin, CZ-level immigrants, population, and average wage for each.

A.5 Shift-Share Construction

(a) National changes: $\Delta M_{o,t} = \text{imm}^{\text{national}}_{o,t} - \text{imm}^{\text{national}}_{o,t-1}$ for $t \in \{1980, 1990, 2000, 2010\}$.

(b) Predicted inflows: $\text{predicted}_{c,o,t} = s_{c,o} \times \Delta M_{o,t}$.

(c) Aggregate: $B_{c,t} = \sum_o \text{predicted}_{c,o,t}$.

(d) Normalize: $\widetilde{B}_{c,t} = B_{c,t}/\text{pop}_{c,1970}$.

A.6 Panel and Analysis

(a) Merge five CZ-level files + shift-share instruments into one dataset.

(b) Compute: $\Delta\text{imm}_{c,t} = (\text{imm}_{c,t} - \text{imm}_{c,t-1})/\text{pop}_{c,1970}$ and $\Delta\text{wage}_{c,t} = \text{wage}_{c,t} - \text{wage}_{c,t-1}$.

(c) Reshape to long panel (CZ $\times$ decade).

(d) Map of the shift-share instrument, first-stage scatter + regression, OLS and IV wage regressions.